Numerical Analysis

Lecture 3: The Solutions of Nonlinear Systems

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Fixed Points for Functions of Several Variables

Nonlinear Systems

A system of nonlinear equations has the form

$$\begin{aligned} f_1(x_1, x_2, \ldots, x_n) &= 0, \\ f_2(x_1, x_2, \ldots, x_n) &= 0, \\ &\vdots \\ f_n(x_1, x_2, \ldots, x_n) &= 0, \end{aligned}$$

which can be represented by

$$\mathbf{F}(\mathbf{x}) = \mathbf{0}$$

The functions $f_1, f_2, \ldots, f_n$ are called the coordinate functions of $\mathbf{F}$.

Outline

1. Vector Norms
2. Fixed Points for Functions of Several Variables
3. Newton's Method for Nonlinear Systems
4. Gradient Descent Techniques

Vector Norms

- Let $\mathbb{R}^n$ denote the set of all n-dimensional column vectors with real-number components.
- To define a distance in $\mathbb{R}^n$ we use the notion of a norm, which is the generalization of the absolute value on $\mathbb{R}$, the set of real numbers.

Definition: Vector Norm

A vector norm on $\mathbb{R}^n$ is a function, $||\cdot||$, from $\mathbb{R}^n$ into $\mathbb{R}$ with the following properties:

1. $||\mathbf{x}|| \geq 0$ for all $\mathbf{x} \in \mathbb{R}^n$
2. $||\mathbf{x}|| = 0$ iff $\mathbf{x} = 0$
3. $||\alpha\mathbf{x}|| = |\alpha|||\mathbf{x}||$ for all $\alpha \in \mathbb{R}$ and $\mathbf{x} \in \mathbb{R}^n$
4. $||\mathbf{x} + \mathbf{y}|| \leq ||\mathbf{x}|| + ||\mathbf{y}||$ for all $||\mathbf{x}||, ||\mathbf{y}|| \in \mathbb{R}^n$.

Vector Norms

Definition: L_1, L_2, and L_∞ Norms

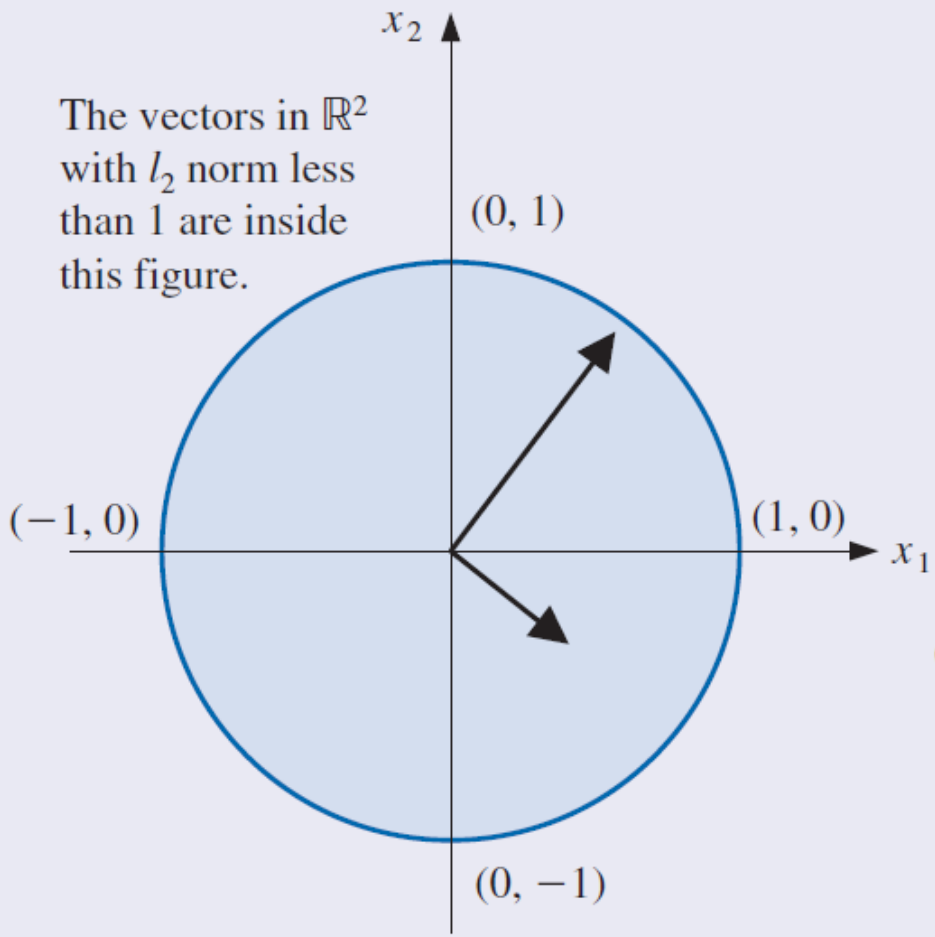
The norms for the vector $\mathbf{x} = (x_1, x_2, \cdots, x_n)^t$ are defined by:

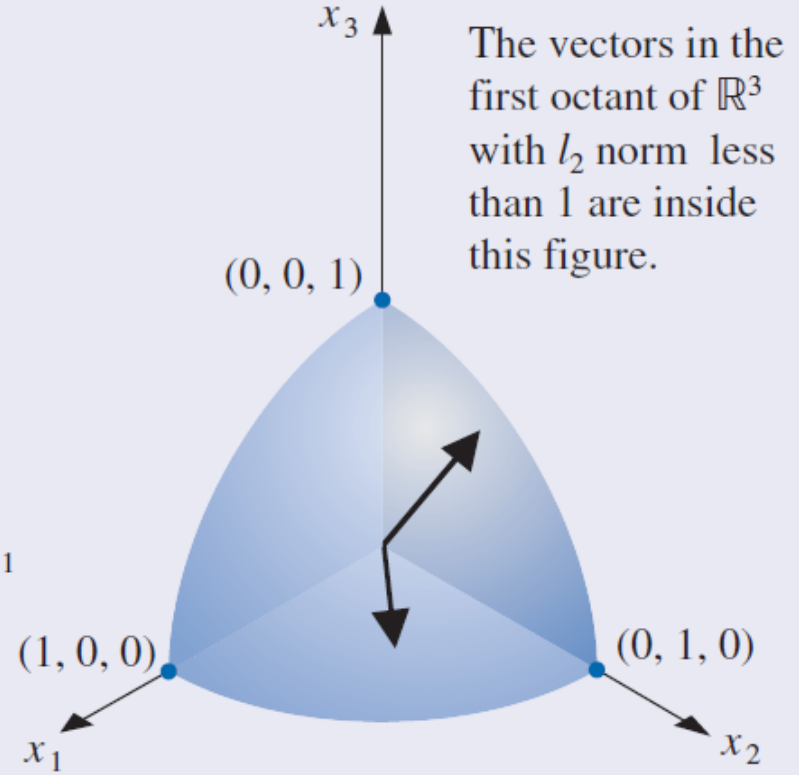
- L_1 Norm
$$||\mathbf{x}||_1 = \sum_{i=1}^{n} |x_i|$$
- L_2 Norm
$$||\mathbf{x}||_2 = \left\{ \sum_{i=1}^{n} x_i^2 \right\}^{1/2}$$
- L_∞ Norm
$$||\mathbf{x}||_\infty = \max_{1 \le i \le n} |x_i|$$

Note that each of these norms reduces to the absolute value in the case $n = 1$.

Vector Norms

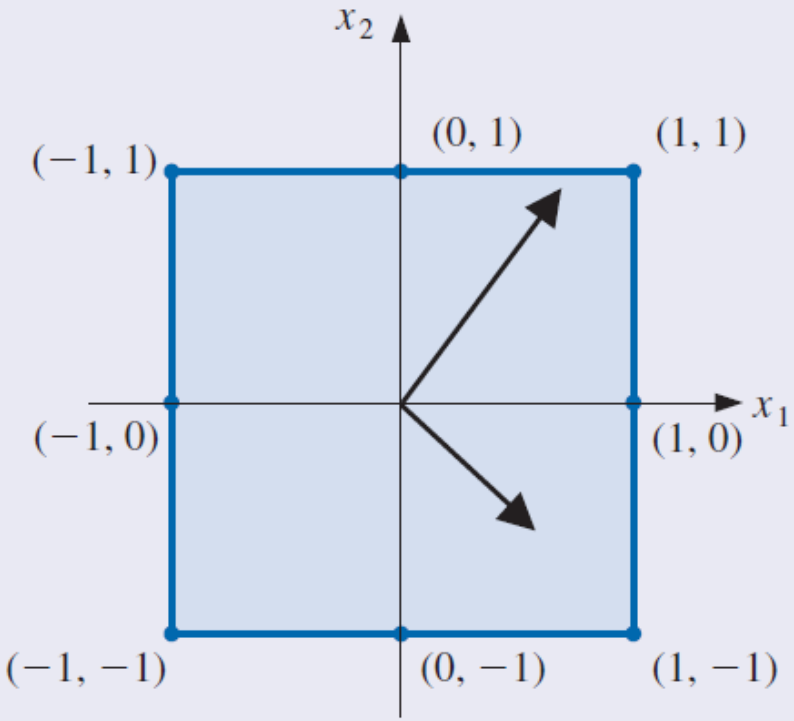
Boundary in $\mathbb{R}^2$ and $\mathbb{R}^3$ with $L_2 < 1$



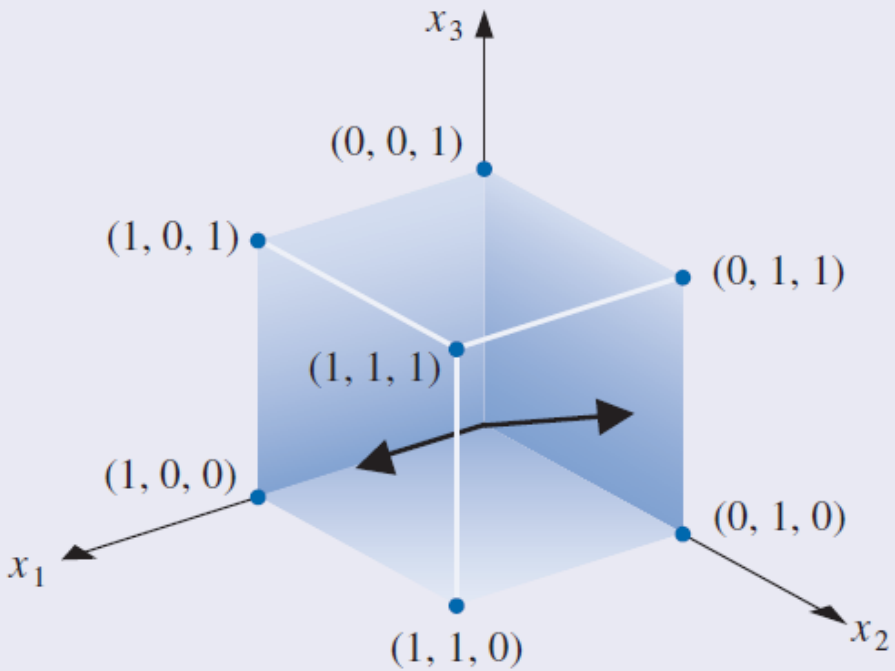


Vector Norms

Boundary in $\mathbb{R}^2$ and $\mathbb{R}^3$ with $L_\infty < 1$



The vectors in $\mathbb{R}^2$ with l_∞ norm less than 1 are inside this figure.



The vectors in the first octant of $\mathbb{R}^3$ with l_∞ norm less than 1 are inside this figure.

Vector Norms

Establishing the Properties of a Vector Norm for L_∞

- It is easy to show that the first three properties of the vector norm hold for L_∞ norm.
- The only property that requires much demonstration is

$$||\mathbf{x} + \mathbf{y}||_\infty \leq ||\mathbf{x}||_\infty + ||\mathbf{y}||_\infty \quad \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$$

Vector Norms

Establishing the Properties of a Vector Norm for L_∞

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- The only property that requires much demonstration is

$$||\mathbf{x}+\mathbf{y}||_\infty \leq ||\mathbf{x}||_\infty + ||\mathbf{y}||_\infty \quad \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$$

- In this case, if $\mathbf{x} = (x_1, x_2, \cdots, x_n)^t$ and $\mathbf{y} = (y_1, y_2, \cdots, y_n)^t$, then

$$\begin{aligned} ||\mathbf{x}+\mathbf{y}||_\infty &= \max_{1\leq i\leq n} |x_i + y_i| \leq \max_{1\leq i\leq n} (|x_i| + |y_i|) \\ &\leq \max_{1\leq i\leq n} |x_i| + \max_{1\leq i\leq n} |y_i| = ||\mathbf{x}||_\infty + ||\mathbf{y}||_\infty \end{aligned}$$

Vector Norms

Establishing the Properties of a Vector Norm for L_2

- It is easy to show that the first three properties of the vector norm hold for L_2 norm.
- But to show that

$$||\mathbf{x}+\mathbf{y}||_2 \leq ||\mathbf{x}||_2 + ||\mathbf{y}||_2 \quad \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$$

we need a famous inequality.

Theorem (Cauchy-Bunyakovsky-Schwarz Inequality for Sums)

For each $\mathbf{x} = (x_1, x_2, \cdots, x_n)^t$ *and* $\mathbf{y} = (y_1, y_2, \cdots, y_n)^t$ *in* $\mathbb{R}^n$,

$$\mathbf{x}^t\mathbf{y} = \sum_{i=1}^{n} x_i y_i \leq \left\{\sum_{i=1}^{n} x_i^2\right\}^{1/2} \left\{\sum_{i=1}^{n} y_i^2\right\}^{1/2} = ||\mathbf{x}||_2 ||\mathbf{y}||_2$$

Vector Norms

Proof (1/2)

- If $\mathbf{y} = \mathbf{0}$ or $\mathbf{x} = \mathbf{0}$, the result is immediate.

Vector Norms

Proof (1/2)

- If $\mathbf{y} = \mathbf{0}$ or $\mathbf{x} = \mathbf{0}$, the result is immediate.
- Suppose $\mathbf{y} \neq \mathbf{0}$ and $\mathbf{x} \neq \mathbf{0}$. Note that, for each $\lambda \in \mathbb{R}$, we have

$$0 \leq ||\mathbf{x} - \lambda \mathbf{y}||_2^2 = \sum_{i=1}^{n} (x_i - \lambda y_i)^2 = \sum_{i=1}^{n} x_i^2 - 2\lambda \sum_{i=1}^{n} x_i y_i + \lambda^2 \sum_{i=1}^{n} y_i^2$$

so that

$$2\lambda \sum_{i=1}^{n} x_i y_i \leq \sum_{i=1}^{n} x_i^2 + \lambda^2 \sum_{i=1}^{n} y_i^2 = ||\mathbf{x}||_2^2 + \lambda^2 ||\mathbf{y}||_2^2$$

However $||\mathbf{x}||_2 > 0$ and $||\mathbf{y}||_2 > 0$, so we can let $\lambda = ||\mathbf{x}||_2 / ||\mathbf{y}||_2$ to give

$$\left(2 \frac{||\mathbf{x}||_2}{||\mathbf{y}||_2} \right) \left(\sum_{i=1}^{n} x_i y_i \right) \leq ||\mathbf{x}||_2^2 + \frac{||\mathbf{x}||_2^2}{||\mathbf{y}||_2^2} ||\mathbf{y}||_2^2 = 2||\mathbf{x}||_2^2$$

Proof (2/2)

Hence

$$2\sum_{i=1}^{n} x_i y_i \leq 2||\mathbf{x}||_2^2 \frac{||\mathbf{y}||_2}{||\mathbf{x}||_2} = 2||\mathbf{x}||_2||\mathbf{y}||_2$$

and

$$\mathbf{x}^t\mathbf{y} = \sum_{i=1}^{n} x_i y_i \leq ||\mathbf{x}||_2||\mathbf{y}||_2 = \left\{\sum_{i=1}^{n} x_i^2\right\}^{1/2} \left\{\sum_{i=1}^{n} y_i^2\right\}^{1/2}$$

Vector Norms

Proof: $||\mathbf{x}+\mathbf{y}||_2 \leq ||\mathbf{x}||_2 + ||\mathbf{y}||_2$

With the Cauchy-Bunyakovsky-Schwarz Inequality, we see that for each $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$,

$$\begin{aligned} ||\mathbf{x}+\mathbf{y}||_2^2 &= \sum_{i=1}^{n} (x_i + y_i)^2 \\ &= \sum_{i=1}^{n} x_i^2 + 2\sum_{i=1}^{n} x_i y_i + \sum_{i=1}^{n} y_i^2 \\ &\leq ||\mathbf{x}||_2^2 + 2||\mathbf{x}||_2||\mathbf{y}||_2 + ||\mathbf{y}||_2^2 \end{aligned}$$

which gives norm property:

$$||\mathbf{x}+\mathbf{y}||_2 \leq \left(||\mathbf{x}||_2^2 + 2||\mathbf{x}||_2||\mathbf{y}||_2 + ||\mathbf{y}||_2^2\right)^{1/2} = ||\mathbf{x}||_2 + ||\mathbf{y}||_2$$

Distance between Vectors in $\mathbb{R}^n$

Definition: Distance between Vectors

If $\mathbf{x} = (x_1, x_2, \cdots, x_n)^t$ and $\mathbf{y} = (y_1, y_2, \cdots, y_n)^t$ are vectors in $\mathbb{R}^n$, the L_2 and L_∞ distances between $\mathbf{x}$ and $\mathbf{y}$ are defined by

$$||\mathbf{x} - \mathbf{y}||_2 = \left\{ \sum_{i=1}^{n} (x_i - y_i)^2 \right\}^{1/2}$$

and

$$||\mathbf{x} - \mathbf{y}||_\infty = \max_{1 \le i \le n} |x_i - y_i|$$

A Limit of a Sequence of Vectors in $\mathbb{R}^n$

Definition: A Limit of a Sequence of Vectors in $\mathbb{R}^n$

A sequence $\{\mathbf{x}^{(k)}\}_{k=1}^{\infty}$ of vectors in $\mathbb{R}^n$ is said to converge to $\mathbf{x}$ with respect to the norm $||\cdot||$ if, given any $\epsilon > 0$, there exists an integer N_ϵ such that

$$||\mathbf{x}^{(k)} - \mathbf{x}|| < \epsilon, \quad \text{for all } k \geq N_\epsilon$$

Theorem

The sequence of vectors $\{\mathbf{x}^{(k)}\}$ *converges to* $\mathbf{x}$ *in* $\mathbb{R}^n$ *with respect to the* L_∞ *norm iff*

$$\lim_{k\to\infty} x_i^{(k)} = x_i$$

for each $i = 1, 2, \cdots, n$.

A Limit of a Sequence of Vectors in $\mathbb{R}^n$

Proof (1/2)

Suppose $\{\mathbf{x}^{(k)}\}$ converges to $\mathbf{x}$ with respect to the L_∞ norm. Given any $\epsilon > 0$, there exists an integer N_ϵ such that for all $k \geq N_\epsilon$,

$$\max_{i=1,2,\cdots,n} |x_i^{(k)} - x_i| = ||\mathbf{x}^{(k)} - \mathbf{x}||_\infty < \epsilon$$

This result implies that $|x_i^{(k)} - x_i| < \epsilon$, for each $i = 1, 2, \cdots, n$, so

$$\lim_{k\to\infty} x_i^{(k)} = x_i$$

for each i.

A Limit of a Sequence of Vectors in $\mathbb{R}^n$

Proof (2/2)

Conversely, suppose that $\lim_{k\to\infty} x_i^{(k)} = x_i$, for every $i = 1, 2, \cdots, n$. For a given $\epsilon > 0$, let $N_{i\epsilon}$ for each i represent an integer with the property that

$$|x_i^{(k)} - x_i| < \epsilon$$

whenever $k \geq N_{i\epsilon}$. Define $N_\epsilon = \max_{i=1,2,\cdots,n} N_{i\epsilon}$. If $k \geq N_\epsilon$, then

$$\max_{i=1,2,\cdots,n} |x_i^{(k)} - x_i| = ||\mathbf{x}^{(k)} - \mathbf{x}||_\infty < \epsilon$$

This implies that $\{\mathbf{x}^{(k)}\}$ converges to $\mathbf{x}$ with respect to the L_∞ norm.

A Limit of a Sequence of Vectors in $\mathbb{R}^n$

Theorem

For each $\mathbf{x} \in \mathbb{R}^n$, $||\mathbf{x}||_\infty \leq ||\mathbf{x}||_2 \leq \sqrt{n}||\mathbf{x}||_\infty$

A Limit of a Sequence of Vectors in $\mathbb{R}^n$

Theorem

For each $\mathbf{x} \in \mathbb{R}^n$, $||\mathbf{x}||_\infty \leq ||\mathbf{x}||_2 \leq \sqrt{n}||\mathbf{x}||_\infty$

Proof

Let x_j be a coordinate of $\mathbf{x}$ such that $||\mathbf{x}||_\infty = \max_{1 \leq i \leq n} |x_i| = |x_j|$.

Then

$$||\mathbf{x}||_\infty^2 = |x_j|^2 \leq \sum_{i=1}^{n} x_i^2 = ||\mathbf{x}||_2^2$$

and

$$||\mathbf{x}||_\infty \leq ||\mathbf{x}||_2$$

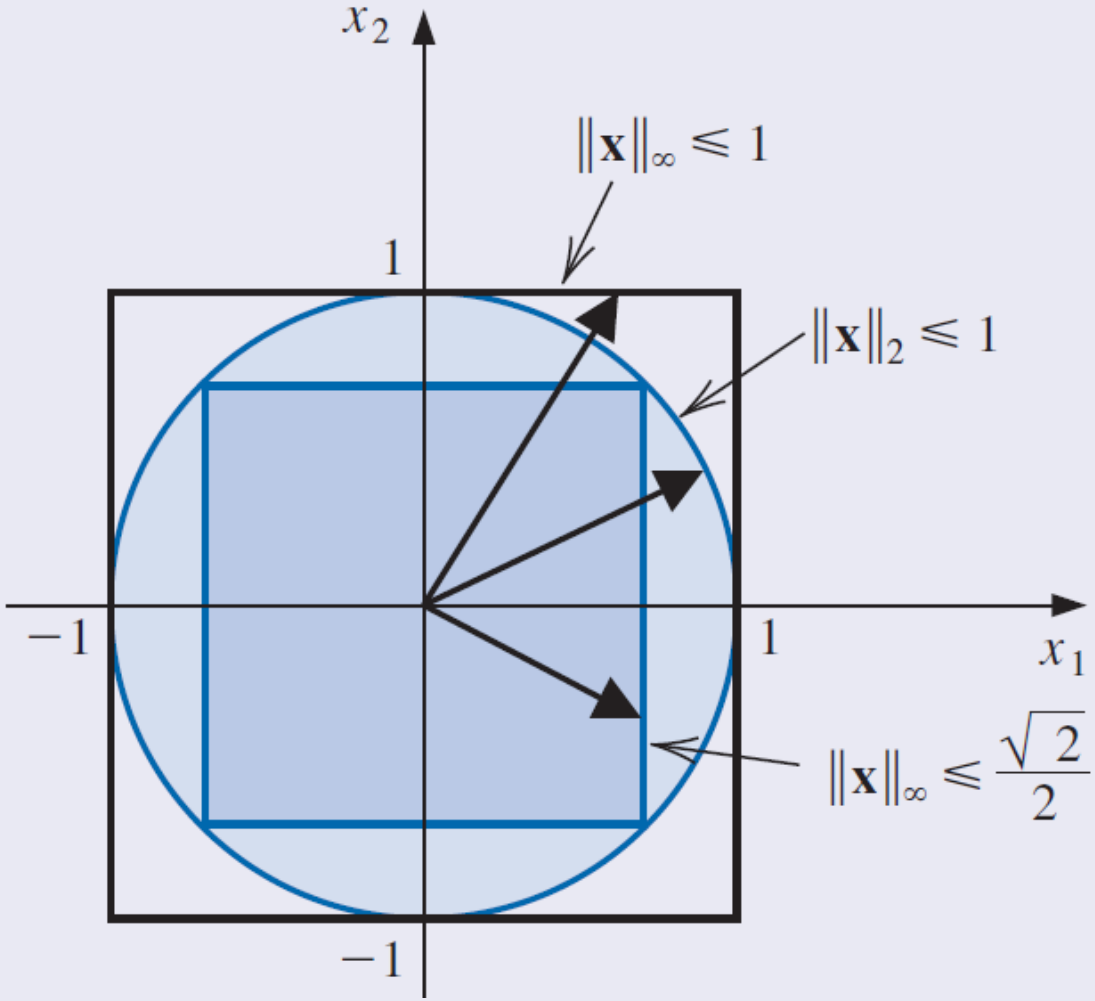
so

$$||\mathbf{x}||_2^2 = \sum_{i=1}^{n} x_i^2 \leq \sum_{i=1}^{n} x_j^2 = nx_j^2 = n||\mathbf{x}||_\infty^2$$

and $||\mathbf{x}||_2 \leq \sqrt{n}||\mathbf{x}||_\infty$.

A Limit of a Sequence of Vectors in $\mathbb{R}^n$

$||\mathbf{x}||_\infty \leq ||\mathbf{x}||_2 \leq \sqrt{n}||\mathbf{x}||_\infty$ Illustration when $n = 2$



A Limit of a Sequence of Vectors in $\mathbb{R}^n$

Theorem

$\| \mathbf{x^{(k)}} - \mathbf{x}\|_\infty < \varepsilon \Rightarrow \| \mathbf{x^{(k)}} - \mathbf{x}\|_\mathbf{2} < \sqrt{\mathbf{n}}\varepsilon$
$\| \mathbf{x^{(k)}} - \mathbf{x}\|_\mathbf{2} < \varepsilon \Rightarrow \| \mathbf{x^{(k)}} - \mathbf{x}\|_\infty < \varepsilon$

Example

Show that

$$\mathbf{x}^{(k)} = (x_1^{(k)}, x_2^{(k)}, x_3^{(k)}, x_4^{(k)})^t = \left(1, 2 + \frac{1}{k}, \frac{3}{k^2}, e^{-k} sin(k)\right)^t$$

- converges to $\mathbf{x} = (1, 2, 0, 0)^t$ with respect to the L_∞ norm.
- It also converges to $\mathbf{x} = (1, 2, 0, 0)^t$ with respect to the L_2 norm.

A Limit of a Sequence of Vectors in $\mathbb{R}^n$

Solution

Given any $\epsilon > 0$, there exists an integer $N(\epsilon/2)$ with the property that

$$||\mathbf{x}^{(k)} - \mathbf{x}||_\infty < \frac{\epsilon}{2},$$

whenever $k \geq N(\epsilon/2)$. It implies that

$$||\mathbf{x}^{(k)} - \mathbf{x}||_2 \leq \sqrt{4}||\mathbf{x}^{(k)} - \mathbf{x}||_\infty \leq 2\frac{\epsilon}{2} = \epsilon$$

when $k \geq N(\epsilon/2)$. So $\{\mathbf{x}^{(k)}\}$ also converges to $\mathbf{x}$ with respect to the L_2 norm.

Outline

1. Vector Norms
2. Fixed Points for Functions of Several Variables
3. Newton's Method for Nonlinear Systems
4. Gradient Descent Techniques

Nonlinear Systems

A system of nonlinear equations has the form

$$\begin{aligned} f_1(x_1, x_2, \ldots, x_n) &= 0, \\ f_2(x_1, x_2, \ldots, x_n) &= 0, \\ &\vdots \\ f_n(x_1, x_2, \ldots, x_n) &= 0, \end{aligned}$$

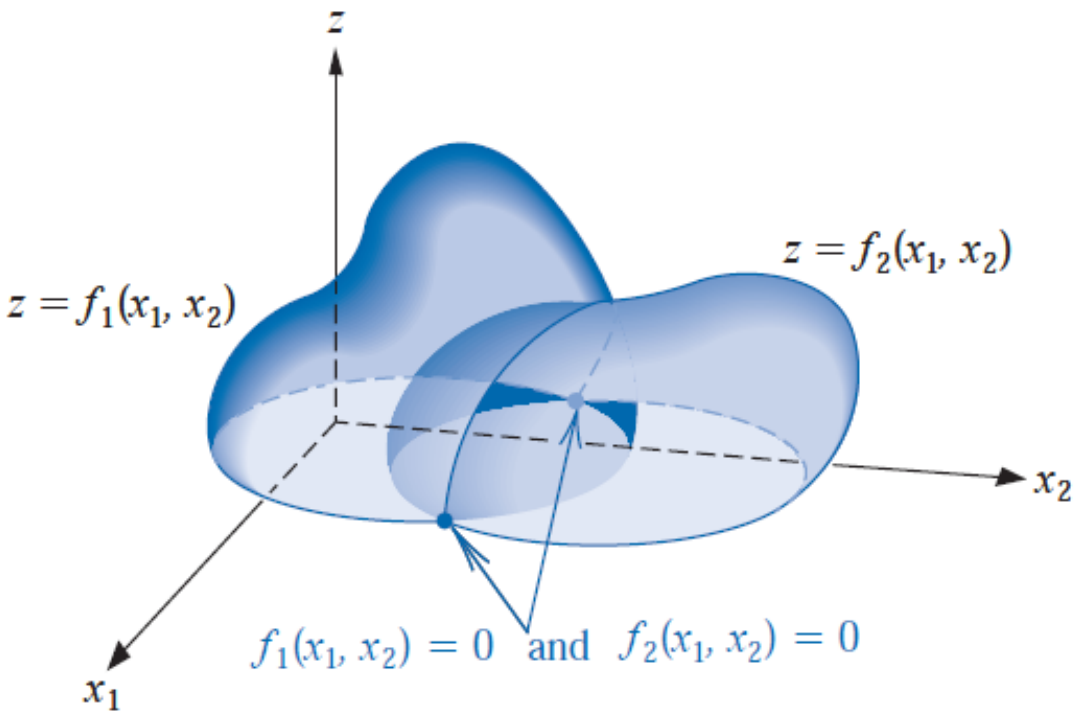
which can be represented by

$$\mathbf{F}(\mathbf{x}) = \mathbf{0}$$

The functions $f_1, f_2, \ldots, f_n$ are called the coordinate functions of $\mathbf{F}$.

Fixed Points for Functions of Several Variables

A System of Two Nonlinear Equations



Fixed Points for Functions of Several Variables

Fixed Points in $\mathbb{R}^n$

A function $\mathbf{G}$ from $\mathbf{D} \in \mathbb{R}^n$ into $\mathbb{R}^n$ has a fixed point at $\mathbf{p} \in D$ if $\mathbf{G}(\mathbf{p}) = \mathbf{p}$.

Theorem (Existence of Fixed Points)

Let $\mathbf{D} = \{(x_1, x_2, \ldots, x_n)^t | a_i \leq x_i \leq b_i, \text{for each } i = 1, 2, \ldots, n\}$ *for some collection of constants* $a_1, a_2, \ldots, a_n$ *and* $b_1, b_2, \ldots, b_n$. *Suppose* $\mathbf{G}$ *is a continuous function from* $\mathbf{D} \subset \mathbb{R}^n$ *into* $\mathbb{R}^n$ *with the property that* $\mathbf{G}(\mathbf{x}) \in \mathbf{D}$ *whenever* $\mathbf{x} \in \mathbf{D}$. *Then* $\mathbf{G}$ *has a fixed point in* $\mathbf{D}$.

Fixed Points for Functions of Several Variables

Theorem (Fixed Point Theorem)

Let $\mathbf{D} = \{(x_1, x_2, \ldots, x_n)^t | a_i \leq x_i \leq b_i, \text{for each} i = 1, 2, \ldots, n\}$ *for some collection of constants* $a_1, a_2, \ldots, a_n$ *and* $b_1, b_2, \ldots, b_n$. *Suppose* $\mathbf{G}$ *is a continuous function from* $\mathbf{D} \in \mathbb{R}^n$ *into* $\mathbb{R}^n$ *with the property that* $\mathbf{G}(\mathbf{x}) \in \mathbf{D}$ *whenever* $\mathbf{x} \in \mathbf{D}$. *Then* $\mathbf{G}$ *has a fixed point in* $\mathbf{D}$.
Moreover, suppose that all the component functions of $\mathbf{G}$ *have continuous partial derivatives and a constant* $K < 1$ *exists with*

$$\left| \frac{\partial g_i(\mathbf{x})}{\partial x_j} \right| \leq \frac{K}{n}, \quad \text{whenever} \quad \mathbf{x} \in \mathbf{D},$$

for each $j = 1, 2, \ldots, n$ *and each component function* g_i. *Then the sequence* $\{\mathbf{x}^{(k)}\}_{k=0}^{\infty}$ *defined by an arbitrarily selected* $\mathbf{x}^{(0)}$ *in* $\mathbf{D}$ *and generated by*

$$\mathbf{x}^{(k)} = \mathbf{G}(\mathbf{x}^{(k-1)}), \quad \text{for each} \quad k \geq 1$$

converges to the unique fixed point $\mathbf{p} \in \mathbf{D}$ *and*

$$||\mathbf{x}^{(k)} - \mathbf{p}||_\infty \leq \frac{K^k}{1 - K} ||\mathbf{x}^{(1)} - \mathbf{x}^{(0)}||_\infty$$

Fixed Points for Functions of Several Variables

Example

Find a solution for the nonlinear system

$$
\begin{aligned}
3x_1 - \cos(x_2 x_3) - \frac{1}{2} &= 0, \\
x_1^2 - 81(x_2 + 0.1)^2 + \sin x_3 + 1.06 &= 0, \\
e^{-x_1 x_2} + 20x_3 + \frac{10\pi - 3}{3} &= 0.
\end{aligned}
$$

Fixed Points for Functions of Several Variables

Example

Find a solution for the nonlinear system

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\begin{aligned}
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e^{-x_1 x_2} + 20x_3 + \frac{10\pi - 3}{3} &= 0.
\end{aligned}
$$

$$
\begin{aligned}
x_1 &= \frac{1}{3}\cos(x_2 x_3) + \frac{1}{6}, \\
x_2 &= \frac{1}{9}\sqrt{x_1^2 + \sin x_3 + 1.06} - 0.1, \\
x_3 &= -\frac{1}{20}e^{-x_1 x_2} - \frac{10\pi - 3}{60}.
\end{aligned}
$$

Fixed Points for Functions of Several Variables

Example

Solution:

$$g_1(x_1, x_2, x_3) = \frac{1}{3}\cos(x_2 x_3) + \frac{1}{6},$$

$$g_2(x_1, x_2, x_3) = \frac{1}{9}\sqrt{x_1^2 + \sin x_3 + 1.06} - 0.1,$$

$$g_3(x_1, x_2, x_3) = -\frac{1}{20}e^{-x_1 x_2} - \frac{10\pi - 3}{60}.$$

k	$x_1^{(k)}$	$x_2^{(k)}$	$x_3^{(k)}$	$\|\mathbf{x}^{(k)} - \mathbf{x}^{(k-1)}\|_\infty$
0	0.10000000	0.10000000	−0.10000000	
1	0.49998333	0.00944115	−0.52310127	0.423
2	0.49999593	0.00002557	−0.52336331	9.4×10^{-3}
3	0.50000000	0.00001234	−0.52359814	2.3×10^{-4}
4	0.50000000	0.00000003	−0.52359847	1.2×10^{-5}
5	0.50000000	0.00000002	−0.52359877	3.1×10^{-7}

Fixed Points for Functions of Several Variables

Acceleration Convergence

Use the latest estimates $x_1^{(k)}, \cdots, x_{i-1}^{(k)}$ instead of $x_1^{(k-1)}, \cdots, x_{i-1}^{(k-1)}$ to compute $x_i^{(k)}$.

$$x_1^{(k)} = \frac{1}{3}\cos\left(x_2^{(k-1)}x_3^{(k-1)}\right) + \frac{1}{6},$$

$$x_2^{(k)} = \frac{1}{9}\sqrt{\left(x_1^{(k)}\right)^2 + \sin x_3^{(k-1)} + 1.06} - 0.1,$$

$$x_3^{(k)} = -\frac{1}{20}e^{-x_1^{(k)}x_2^{(k)}} - \frac{10\pi - 3}{60}.$$

k	$x_1^{(k)}$	$x_2^{(k)}$	$x_3^{(k)}$	$\|\mathbf{x}^{(k)} - \mathbf{x}^{(k-1)}\|_\infty$
0	0.10000000	0.10000000	−0.10000000	
1	0.49998333	0.02222979	−0.52304613	0.423
2	0.49997747	0.00002815	−0.52359807	2.2×10^{-2}
3	0.50000000	0.00000004	−0.52359877	2.8×10^{-5}
4	0.50000000	0.00000000	−0.52359877	3.8×10^{-8}

Outline

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3. Newton's Method for Nonlinear Systems
4. Gradient Descent Techniques

Newton's Method for Nonlinear Systems

The fixed-point method for nonlinear equations

$$g(x) = x - \phi(x)f(x).$$

The fixed-point method for nonlinear systems

$$G(\mathbf{x}) = \mathbf{x} - \mathbf{A}(\mathbf{x})^{-1}\mathbf{F}(\mathbf{x}),$$

where

$$\mathbf{A}(\mathbf{x}) = \begin{bmatrix} a_{11}(\mathbf{x}) & a_{12}(\mathbf{x}) & \dots & a_{1n}(\mathbf{x}) \\ a_{21}(\mathbf{x}) & a_{22}(\mathbf{x}) & \dots & a_{2n}(\mathbf{x}) \\ \dots & \dots & \dots & \dots \\ a_{n1}(\mathbf{x}) & a_{n1}(\mathbf{x}) & \dots & a_{nn}(\mathbf{x}) \end{bmatrix}$$

Newton's Method for Nonlinear Systems

Theorem (Convergence)

Let $\mathbf{p}$ *be a solution of* $\mathbf{G}(\mathbf{x}) = \mathbf{x}$. *Suppose a number* $\delta > 0$ *exists with*

1. $\partial g_i/\partial x_j$ *is continuous on* $N_\delta = \{\mathbf{x} | ||\mathbf{x} - \mathbf{p}|| < \delta\}$, *for each* $i = 1, 2, \cdots, n$ *and* $j = 1, 2, \cdots, n$;
2. $\partial^2 g_i(\mathbf{x})/(\partial x_j \partial x_k)$ *is continuous, and* $|\partial^2 g_i(\mathbf{x})/(\partial x_j \partial x_k)| \leq M$ *for some constant* M, *whenever* $\mathbf{x} \in N_\delta$, *for each* $i = 1, 2, \cdots, n$, $j = 1, 2, \cdots, n$, *and* $k = 1, 2, \cdots, n$;
3. $\partial g_i(\mathbf{p})/\partial x_k = 0$, *for each* $i = 1, 2, \cdots, n$ *and* $k = 1, 2, \cdots, n$.

Then a number $\hat{\delta} \leq \delta$ *exists such that the sequence generated by* $\mathbf{x}^{(k)} = \mathbf{G}(\mathbf{x}^{(k-1)})$ *converges quadratically to* $\mathbf{p}$ *for any choice of* $\mathbf{x}^{(0)}$, *provided that* $||\mathbf{x}^{(0)} - \mathbf{p}|| < \hat{\delta}$. *Moreover,*

$$||\mathbf{x}^{(k)} - \mathbf{p}||_\infty \leq \frac{n^2 M}{2} ||\mathbf{x}^{(k-1)} - \mathbf{p}||_\infty^2, \text{for each } k \geq 1$$

Newton's Method for Nonlinear Systems

Construction of the Matrix

Let $b_{ij}(\mathbf{x})$ denote the entry of $\mathbf{A}(\mathbf{x})^{-1}$, we have

$$g_i(\mathbf{x}) = x_i - \sum_{j=1}^{n} b_{ij}(\mathbf{x}) f_j(\mathbf{x})$$

$$\frac{\partial g_i}{\partial x_k}(\mathbf{x}) = \begin{cases} 1 - \sum_{j=1}^{n} \left(b_{ij}(\mathbf{x}) \frac{\partial f_i}{\partial x_k}(\mathbf{x}) + \frac{\partial b_{ij}}{\partial x_k}(\mathbf{x}) f_j(\mathbf{x}) \right), \text{if } i = k, \\ - \sum_{j=1}^{n} \left(b_{ij}(\mathbf{x}) \frac{\partial f_i}{\partial x_k}(\mathbf{x}) + \frac{\partial b_{ij}}{\partial x_k}(\mathbf{x}) f_j(\mathbf{x}) \right), \text{if } i \neq k. \end{cases}$$

This means that for $i = k$, $\sum_{j=1}^{n} b_{ij}(\mathbf{p}) \frac{\partial f_j}{\partial x_i}(\mathbf{p}) = 1$

and for $i \neq k$, $\sum_{j=1}^{n} b_{ij}(\mathbf{p}) \frac{\partial f_j}{\partial x_i}(\mathbf{p}) = 0$

Newton's Method for Nonlinear Systems

$$J(\mathbf{x}) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1}(\mathbf{x}) & \frac{\partial f_1}{\partial x_2}(\mathbf{x}) & \cdots & \frac{\partial f_1}{\partial x_n}(\mathbf{x}) \\ \frac{\partial f_2}{\partial x_1}(\mathbf{x}) & \frac{\partial f_2}{\partial x_2}(\mathbf{x}) & \cdots & \frac{\partial f_2}{\partial x_n}(\mathbf{x}) \\ \vdots & \vdots & & \vdots \\ \frac{\partial f_n}{\partial x_1}(\mathbf{x}) & \frac{\partial f_n}{\partial x_2}(\mathbf{x}) & \cdots & \frac{\partial f_n}{\partial x_n}(\mathbf{x}) \end{bmatrix}.$$

The Jacobian matrix

Then, $\mathbf{A}(\mathbf{p})^{-1}\mathbf{J}(\mathbf{p}) = \mathbf{I}$. Therefore, $\mathbf{A}(\mathbf{p}) = \mathbf{J}(\mathbf{p})$.

Newton's Method for Nonlinear Systems

$$x^{(k)} = x^{(k-1)} - \frac{1}{f'(x^{(k-1)})} f(x^{(k-1)})$$

Newton's Method for Nonlinear Systems

$$\mathbf{x}^{(k)} = \mathbf{x}^{(k-1)} - \mathbf{J}(\mathbf{x}^{(k-1)})^{-1}\mathbf{F}(\mathbf{x}^{(k-1)}),$$

where $\mathbf{J}(\mathbf{x})$ is the Jacobian matrix

$$J(\mathbf{x}) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1}(\mathbf{x}) & \frac{\partial f_1}{\partial x_2}(\mathbf{x}) & \cdots & \frac{\partial f_1}{\partial x_n}(\mathbf{x}) \\ \frac{\partial f_2}{\partial x_1}(\mathbf{x}) & \frac{\partial f_2}{\partial x_2}(\mathbf{x}) & \cdots & \frac{\partial f_2}{\partial x_n}(\mathbf{x}) \\ \vdots & \vdots & & \vdots \\ \frac{\partial f_n}{\partial x_1}(\mathbf{x}) & \frac{\partial f_n}{\partial x_2}(\mathbf{x}) & \cdots & \frac{\partial f_n}{\partial x_n}(\mathbf{x}) \end{bmatrix}.$$

Newton's Method for Nonlinear Systems

$$\mathbf{x}^{(k)} = \mathbf{x}^{(k-1)} - \mathbf{J}(\mathbf{x}^{(k-1)})^{-1}\mathbf{F}(\mathbf{x}^{(k-1)}),$$

Remarks

- A weakness in Newton's method arises from the need to compute and invert the matrix $\mathbf{J}(\mathbf{x})$ at each step.
- In practice, explicit computation of $\mathbf{J}(\mathbf{x})^{-1}$ is avoided by performing the operation in a two-step manner.
  1. A vector $\mathbf{y}$ is found that satisfies $\mathbf{J}(\mathbf{x}^{(k-1)})\mathbf{y} = -\mathbf{F}(\mathbf{x}^{(k-1)})$
  2. The new approximation, $\mathbf{x}^{(k)}$, is obtained by adding $\mathbf{y}$ to $\mathbf{x}^{(k-1)}$.

Newton's Method for Nonlinear Systems

Algorithm

INPUT number n of equations and unknowns; initial approximation $\mathbf{x} = (x_1, \ldots, x_n)^t$, tolerance TOL; maximum number of iterations N.
OUTPUT approximate solution $\mathbf{x} = (x_1, \ldots, x_n)^t$ or a message that the number of iterations was exceeded.

Step 1 Set $k = 1$.

Step 2 While $(k \leq N)$ do Steps 3–7.

Step 3 Calculate $\mathbf{F}(\mathbf{x})$ and $J(\mathbf{x})$, where $J(\mathbf{x})_{i,j} = (\partial f_i(\mathbf{x})/\partial x_j)$ for $1 \leq i, j \leq n$.

Step 4 Solve the $n \times n$ linear system $J(\mathbf{x})\mathbf{y} = -\mathbf{F}(\mathbf{x})$.

Step 5 Set $\mathbf{x} = \mathbf{x} + \mathbf{y}$.

Step 6 If $||\mathbf{y}|| < TOL$ then OUTPUT $(\mathbf{x})$;
(*The procedure was successful.*)
STOP.

Step 7 Set $k = k + 1$.

Step 8 OUTPUT ('Maximum number of iterations exceeded');
(*The procedure was unsuccessful.*)
STOP.

Newton's Method for Nonlinear Systems

Example

Find a solution for the nonlinear system

$$3x_1 - \cos(x_2x_3) - \frac{1}{2} = 0,$$
$$x_1^2 - 81(x_2 + 0.1)^2 + \sin x_3 + 1.06 = 0,$$
$$e^{-x_1x_2} + 20x_3 + \frac{10\pi - 3}{3} = 0.$$

k	$x_1^{(k)}$	$x_2^{(k)}$	$x_3^{(k)}$	$\|\mathbf{x}^{(k)} - \mathbf{x}^{(k-1)}\|_\infty$
0	0.1000000000	0.1000000000	−0.1000000000	
1	0.4998696728	0.0194668485	−0.5215204718	0.4215204718
2	0.5000142403	0.0015885914	−0.5235569638	1.788×10^{-2}
3	0.5000000113	0.0000124448	−0.5235984500	1.576×10^{-3}
4	0.5000000000	8.516×10^{-10}	−0.5235987755	1.244×10^{-5}
5	0.5000000000	-1.375×10^{-11}	−0.5235987756	8.654×10^{-10}

Newton's Method for Nonlinear Systems

Remarks

- The advantage of the Newton's method for solving systems of nonlinear equations is its speed of convergence once a sufficiently accurate approximation is known.
- A weakness of this method is that an accurate initial approximation to the solution is needed to ensure convergence.

Outline

1. Vector Norms
2. Fixed Points for Functions of Several Variables
3. Newton's Method for Nonlinear Systems
4. Gradient Descent Techniques

Gradient Descent Techniques

Gradient Descent

- A.K.A Steepest Descent method, converges only linearly to the solution, but it will usually converge even for poor initial approximations.
- The method of Steepest Descent determines a local minimum for a multivariable function.

Gradient Descent Techniques

Gradient Descent Method

A system of nonlinear equations has the form

$$\begin{aligned} f_1(x_1, x_2, \ldots, x_n) &= 0, \\ f_2(x_1, x_2, \ldots, x_n) &= 0, \\ &\vdots \\ f_n(x_1, x_2, \ldots, x_n) &= 0. \end{aligned}$$

Then the following function has the minimal value of 0:

$$g(x_1, x_2, \cdots, x_n) = \sum_{i=1}^{n} f_i^2(x_1, x_2, \cdots, x_n)$$

Gradient Descent Techniques

Gradient Descent Method

The method of Steepest Descent for finding a local minimum for an arbitrary function **g** from $\mathbb{R}^n$ into $\mathbb{R}$ can be intuitively described as follows:

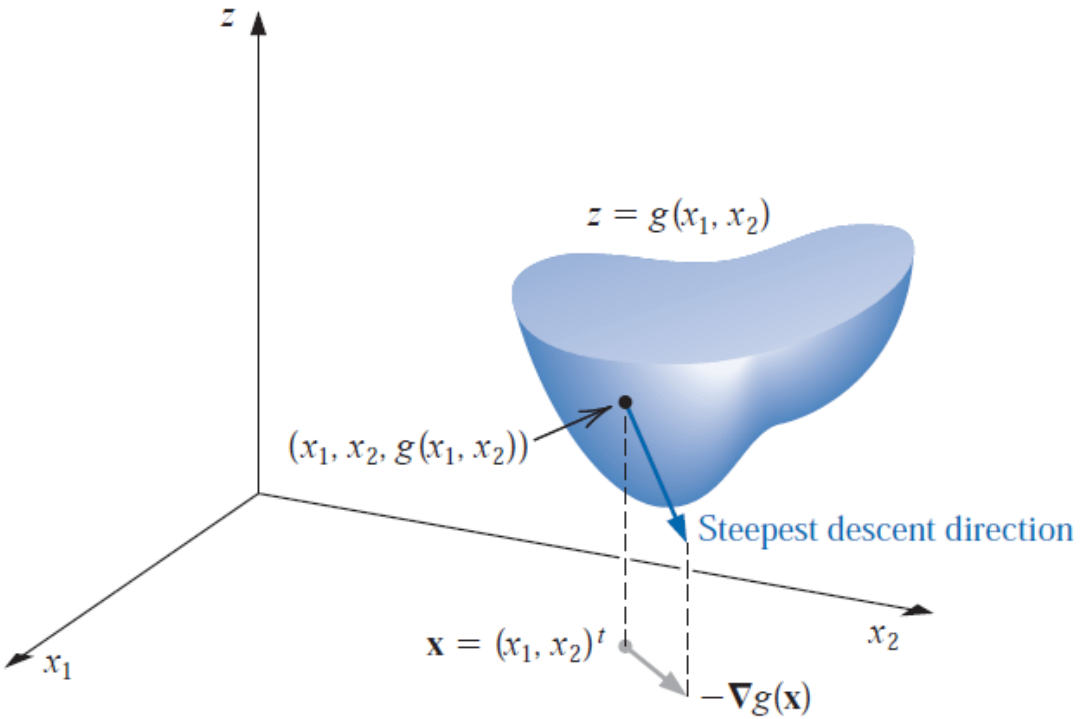
1. Evaluate **g** at an initial approximation $\mathbf{x}^{(0)}$;
2. Determine a direction from $\mathbf{x}^{(0)}$ that results in a decrease in the value of **g**;
3. Move an appropriate amount in this direction and call the new value $\mathbf{x}^{(1)}$;
4. Repeat steps 1 through 3 with $\mathbf{x}^{(0)}$ replaced by $\mathbf{x}^{(1)}$.

Gradient Descent Techniques

The Gradient of a Function

For $g : \mathbb{R}^n \to \mathbb{R}$, the gradient of g at $\mathbf{x} = (x_1, x_2, \cdots, x_n)^t$ is denoted $\nabla g(\mathbf{x})$ and defined by

$$\nabla g(\mathbf{x}) = \left(\frac{\partial g}{\partial x_1}(\mathbf{x}), \frac{\partial g}{\partial x_2}(\mathbf{x}), \cdots, \frac{\partial g}{\partial x_n}(\mathbf{x}) \right)^t.$$



Gradient Descent Techniques

Gradient Descent Method

$$\mathbf{x}^{(k)} = \mathbf{x}^{(k-1)} - \alpha \nabla \mathbf{g}(\mathbf{x}^{(k-1)}),$$

where α is the step size.

Gradient Descent Techniques

Example

Using gradient descent method to find a solution for the nonlinear system

$$
\begin{aligned}
3x_1 - \cos(x_2 x_3) - \frac{1}{2} &= 0, \\
x_1^2 - 81(x_2 + 0.1)^2 + \sin x_3 + 1.06 &= 0, \\
e^{-x_1 x_2} + 20x_3 + \frac{10\pi - 3}{3} &= 0.
\end{aligned}
$$

Gradient Descent Techniques

Example

Using gradient descent method to find a solution for the nonlinear system

$$3x_1 - \cos(x_2x_3) - \frac{1}{2} = 0,$$

$$x_1^2 - 81(x_2 + 0.1)^2 + \sin x_3 + 1.06 = 0,$$

$$e^{-x_1x_2} + 20x_3 + \frac{10\pi - 3}{3} = 0.$$

Solution Let $g(x_1, x_2, x_3) = [f_1(x_1, x_2, x_3)]^2 + [f_2(x_1, x_2, x_3)]^2 + [f_3(x_1, x_2, x_3)]^2$.

k	$x_1^{(k)}$	$x_2^{(k)}$	$x_3^{(k)}$	$g(x_1^{(k)}, x_2^{(k)}, x_3^{(k)})$
2	0.137860	−0.205453	−0.522059	1.27406
3	0.266959	0.00551102	−0.558494	1.06813
4	0.272734	−0.00811751	−0.522006	0.468309
5	0.308689	−0.0204026	−0.533112	0.381087
6	0.314308	−0.0147046	−0.520923	0.318837
7	0.324267	−0.00852549	−0.528431	0.287024

Gradient Descent vs. Newton's Method

Gradient Descent vs. Newton's Method

- Newton's method is to find a root $\mathbf{F}(\mathbf{x}) = 0$

$$\mathbf{x}^{(k)} = \mathbf{x}^{(k-1)} - \mathbf{J}(\mathbf{x}^{(k-1)})^{-1}\mathbf{F}(\mathbf{x}^{(k-1)}),$$

- Gradient descent is to find a local minimum $g(\mathbf{x}) = ||\mathbf{F}(\mathbf{x})||_2^2$

$$\mathbf{x}^{(k)} = \mathbf{x}^{(k-1)} - \alpha \nabla \mathbf{g}(\mathbf{x}^{(k-1)}).$$

Assignments

- Reading Assignment: Chap 10
- Homework 2.